

Marketing Campaign Performance & ROI Optimization

Real-World Business Problem Statement | Data Analytics Portfolio Project

1. Business Context

A growing company is running hundreds of marketing campaigns across multiple channels. The marketing team is investing significant amounts of money in these campaigns but does not have a reliable view of which campaigns and channels are generating the best business results.

The existing campaign data has been collected from multiple sources and contains missing values, inconsistent formats, duplicate fields, and other data-quality issues. These problems make it difficult for management to trust campaign reports and make informed budget decisions.

2. Business Problem

The company wants the Data Analytics team to clean, validate, analyze, and visualize its marketing campaign data to identify high-performing and underperforming campaigns, evaluate channel efficiency, and provide recommendations for optimizing marketing spend and improving conversions.

3. Your Role

You are working as a Junior Data Analyst / Marketing Data Analyst. Your responsibility is to transform the raw campaign data into reliable insights that can support marketing management decisions.

- Assess and clean the raw campaign data.
- Identify and resolve missing, duplicate, inconsistent, and invalid values.
- Standardize dates, channels, spend, and campaign status.
- Analyze campaign performance across marketing channels.
- Identify high-performing and underperforming campaigns.
- Calculate important marketing KPIs.
- Evaluate whether increased marketing spend is associated with higher conversions.
- Identify opportunities to optimize the marketing budget.
- Build an executive dashboard for the Marketing Manager/CMO.
- Provide data-driven recommendations for future campaigns.

4. Key Business Questions

Which marketing channels perform best? Compare impressions, clicks, conversions, CTR, conversion rate, and efficiency across channels.

Which channels use marketing spend most efficiently? Evaluate CPC and cost per conversion alongside total spend and conversions.

Which campaigns are wasting money? Identify campaigns with high spend but low clicks, conversions, or poor conversion efficiency.

Which campaigns should receive more budget? Find campaigns with strong conversion performance and relatively efficient acquisition costs.

Does spending more produce more conversions? Analyze the relationship between campaign spend and conversions.

How does campaign performance change over time? Analyze monthly or time-based trends in spend, clicks, conversions, and efficiency.

Are active campaigns performing better? Compare active and inactive campaigns using relevant performance metrics.

Which campaign tags perform best? Compare campaign categories/tags based on spend, conversions, conversion rate, and efficiency.